



Autism spectrum disorders

Autism spectrum disorders (ASD) is a term used to describe a group of developmental problems, all of which are characterized by communication and behavioral difficulties. The word "spectrum" refers to the wide variety of types and severity levels of symptoms experienced by people with ASD.

People who have ASD find it hard to interact and communicate with others. They also tend to have restricted interests and repetitive behaviors and are often more or less sensitive than others to light, sound, or temperature. This causes them to retreat into themselves.

ASD occurs in people at all levels of intellectual ability and is most often diagnosed in the first two years of life. It is a lifelong condition. Physical symptoms may include repetitive body movements, such as pacing, rocking, or hand flapping.

Communication problems

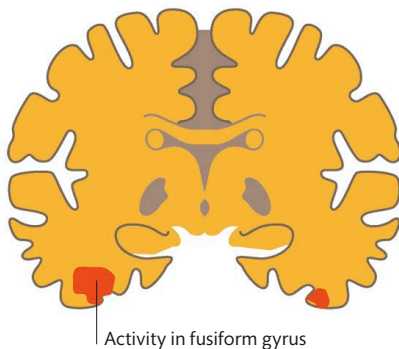
Children with ASD may have language difficulties, and some start talking relatively late in life. Their tone of voice might be very flat, very fast, or singsong. About 40 percent of children with ASD don't talk at all, and between 25 and 30 percent develop some language skills during infancy but then lose them later in life.

SYMPTOMS OF AUTISM SPECTRUM DISORDERS	
SYMPTOM	DESCRIPTION
Social communication	ASD affects social communication because the development of language is impaired. Verbal and nonverbal social communication problems include difficulties interpreting social situations, identifying social cues, and blunt or inappropriate conversational interactions.
Repetitive behavior	People with ASD often engage in repeated activities, such as hand flapping or body rocking, or may harm themselves by continuous biting or skin picking. They may also exhibit body twirling or other complex body movements, along with rituals such as counting or arranging objects.
Focused interests	Those with autism often think in very black-and-white terms, with an intense focus on specific interests or obsessions. These can range from spinning objects to collecting birthdates or identifying flight paths.
Sensory sensitivity	Some type of sensory processing problem is usually (although not always) related to a diagnosis of ASD. Those affected may be over- or undersensitive and experience difficulties with smell, taste, sight, hearing, touch, balance, eye movement, and body awareness.

High-functioning adults with ASD may be successful in academic fields yet have difficulty with practical and social skills, such as understanding social cues. Most seem blunt, cannot lie, and may focus obsessively on one aspect of life, such as cleanliness.

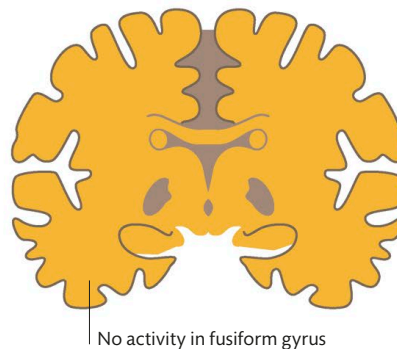
Social awkwardness is usually accompanied by social anxiety. Other symptoms of ASD include a highly acute awareness of noise, smell, touch, or light, and extreme food preferences.

ASD sufferers who have intellectual disabilities may show a high aptitude in other areas such as having a photographic memory or numerical ability; however, sometimes the disability is so profound that those affected cannot speak meaningfully, engage in self-harm, and need daily care.



Activity in fusiform gyrus

NORMAL BRAIN



No activity in fusiform gyrus

AUTISTIC BRAIN

ASD and normal brain comparisons

Those with ASD find it hard to process faces. In a nonautistic person, activity shows in the temporal lobe's fusiform gyrus, where recognition occurs. In the autistic brain, there is no such corresponding activity.